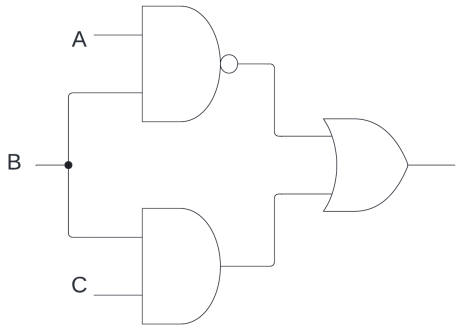
Drew University - CSCI 260 - Fall 2024
HW2 - Digital Logic

HW Info.: Due on 09/27/2024 (F) by 11:59pm, 34 points
Submission: Submit as a single PDF file on Moodle.
Note: Please follow Drew University's Standard of Academic Integrity and refer to the course syllabus for late submission policy.

Note: For the following problems, if you are asked to provide a truth table, you need to **show all your work**, meaning that you need to include and fill all the necessary intermediate columns.

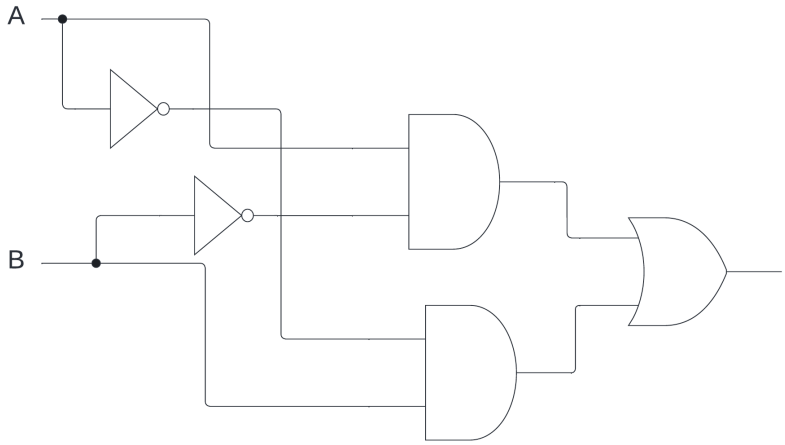
1. (12 points) For the following logical expressions, first give their corresponding gate-level circuits (i.e., diagrams with logic gates connected by wires) and then use truth tables to evaluate them. **Show all your work**.

   (a) NOT(A AND B) OR (C AND B)



A	B	C	X = A NAND B	Y = C AND B	X OR Y
0	0	0	1	0	1
0	0	1	1	0	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	1	0	1
1	0	1	1	0	1
1	1	0	0	0	0
1	1	1	0	1	1

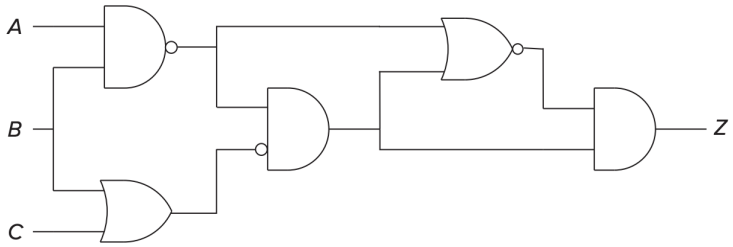
   (b) (A AND NOT(B)) OR (NOT(A) AND B)



A	B	NOT(B)	NOT(A)	X = A AND NOT(B)	Y = NOT(A) AND B	X OR Y
0	0	1	1	0	0	0
0	1	0	1	0	1	1
1	0	1	0	1	0	1
1	1	0	0	0	0	0

   [**Note**: This is A XOR B.]

2. (12 points) Given the following gate-level logical circuit, give the equivalence logical expression of the output Z and then use a truth table to evaluate the circuit. **Show all your work**.



Ans:

- Top-left gate: V = A NAND B
- Bottom-left gate: W= B OR C
- Middle gate: X = V AND NOT(W)
- Top-right gate: Y = V NOR X
- Right-most gate: Z = Y AND X

=((A NAND B) NOR ((A NAND B) AND NOT(B OR C))) AND ((A NAND B) AND NOT(B OR C))

A	B	C	V	W	NOT(W)	X	Y	Z
0	0	0	1	0	1	1	0	0
0	0	1	1	1	0	0	0	0
0	1	0	1	1	0	0	0	0
0	1	1	1	1	0	0	0	0
1	0	0	1	0	1	1	0	0
1	0	1	1	1	0	0	0	0
1	1	0	0	1	0	0	1	0
1	1	1	0	1	0	0	1	0

[**Note**: For those of you who've taken MATH 220 Discrete Math, you can try to simplify the expression of Z to simply a 0, for "fun".]

3. (4 points) Answer the following questions regarding the computer memory structure.

   (a) Given a memory that has a 5-byte addressability and needs 4 bits to access all the memory locations, how many bits of storage does the memory contain at maximum?

   Ans: Since we need 4 bits to access all the memory locations, there are $2^4 = 16$ total locations. Addressability is 5-byte means there are 5 bytes per location, or $5 \times 8 = 40$ bits per location. Therefore, in total we have $16 \times 40 = 640$ bits in the memory.

   (b) Given a memory that is addressed by 17 bits and is 16-bit addressable, how many bytes of storage does the memory contain at maximum?

   Ans: Similarly, there are 2^{17} total locations, with each location 16-bit, or 2-byte, wide. Therefore, in total, we have $2^{17} \times 2 = 2^{18} = 262,144$ bytes in the memory.

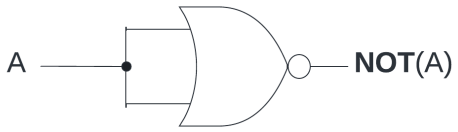
   [**Note**: 2^{20} bytes is 1 MB; hence, this is 0.25 MB.]

4. (6 points) Show that the NOR gate, by itself, is logically complete. To do so, construct logical circuits (i.e., diagrams) that perform the NOT, AND, and OR operations by only using the NOR gate. **Show all your work**.

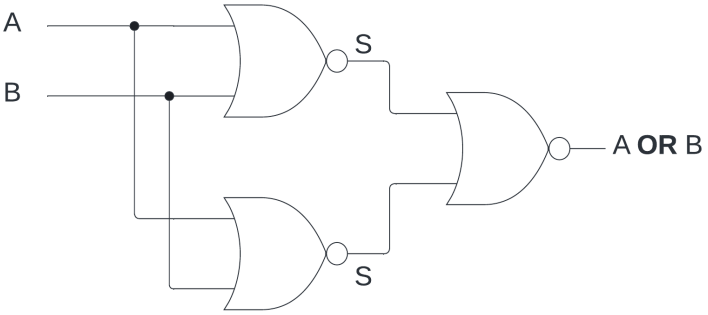
   Ans: The logical gates are defined as follows with their circuits and truth tables to verify.

   - NOT(A) = A NOR A.
   - Let S = A NOR B, then A OR B = S NOR S.
   - Given NOT and OR, apply De Morgan's law on NOT(NOT(A) OR NOT(B)), or ((A NOR A) NOR (B NOR B)) to acquire A AND B.

A	A NOR A
0	1
1	0



A	B	S = A NOR B	S NOR S
0	0	1	0
0	1	0	1
1	0	0	1
1	1	0	1



A	B	NOT A	NOT B	NOT(A) NOR NOT(B)
0	0	1	1	0
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1

